Victor Heorhiadi

Staff-level Engineer/Scientist — Networking, Distributed Systems & Formal Methods

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github.com/progwriter | scholar.google.com/citations?user=9Eu_XLMAAAAJ

Summary

Staff-level engineer/scientist with a decade of experience spanning networking, distributed systems and formal methods. At AWS, architected core control-plane and routing infrastructure operating across dozens of datacenters and hundreds of thousands of devices, and applied formal and neurosymbolic verification techniques to guardrail AI-driven workflows for large enterprises. Previously a core developer of Batfish, an open-source network configuration analysis tool. Holds a PhD in Computer Science from UNC Chapel Hill.

Skills

Systems/Infrastructure: Routing protocols, ML networking infrastructure, network control planes, network operations

Formal methods & Verification: SMT and solvers (e.g. Z3), neurosymbolic AI guardrails

Practices: Property-based testing, model checking, fault injection, network simulation

Languages: Rust, Python, Java/Kotlin

Experience

Sr. Applied Scientist, Amazon Web Services – Seattle, WA 2023 – present

- Architected the "network compiler" for AIDN (AWS Intent-Driven Networking), AWS's next-generation network control plane, now operating across dozens of datacenters and hundreds of thousands of network devices, including AI/ML UltraClusters
- Contributed to the implementation of SIDR (Scalable Intent-Driven Routing), a Rust-based link-state routing protocol for ultra-large datacenters, and led the correctness effort using model checking and property-based testing that enabled an accelerated multi-datacenter rollout
- Contributed to AWS Automated Reasoning Checks, a neurosymbolic AI guardrail enabling compliance verification of AI-driven workflows. Used by large enterprises for proving correctness of financial workflows

Applied Scientist, Amazon Web Services – Seattle, WA 2021 – 2023

- Designed and implemented an internal formal-verification service that proves correctness of network access control lists, auditing security policy across AWS's production networking infrastructure

Software Engineer, Intentionet – Seattle, WA 2017 – 2021

- Core developer of Batfish, an open-source network configuration analysis tool. Focus on routing protocols and control plane simulation
- Led development of Batfish's routing simulation engine, adding support for numerous routing protocols and features that enabled onboarding networks for several Fortune 500 companies

Graduate Research Assistant, University of North Carolina at Chapel Hill – Chapel Hill, NC 2011 – 2017

- Researched Software-Defined Networking (SDN) controllers, applying global optimization techniques (linear programming) to compute optimal traffic paths and program SDN switches

Research Internship, IBM Research – Yorktown Heights, NY June 2015 – Aug 2015

- Built a fault-injection framework for testing microservice applications

Education

University of North Carolina at Chapel Hill, PhD in Computer Science – Chapel Hill, NC 2017

University of North Carolina at Chapel Hill, MS in Computer Science – Chapel Hill, NC 2013

University of Minnesota Twin Cities, BS in Computer Science – Minneapolis, MN 2011

Selected Publications

A Neurosymbolic Approach to Natural Language Formalization and Verification 2026
Chenyang An, et al.
10.1007/978-3-032-32526-6_28 (38th International Conference on Computer Aided Verification (CAV 2026))

Lessons from the evolution of the Batfish configuration analysis tool 2023
Matt Brown, et al.
10.1145/3603269.3604866 (ACM SIGCOMM 2023 Conference (SIGCOMM '23))

Intent-driven Composition of Resource-management SDN Applications 2018
Victor Heorhiadi, Sanjay Chandrasekaran, et al.
10.1145/3281411.3281431 (ACM CoNEXT 2018 (CoNEXT '18))

Gremlin: Systematic Resilience Testing of Microservices June 2016
Victor Heorhiadi, Shriram Rajagopalan, et al.
(IEEE ICDCS 2016 (ICDCS '16))

Simplifying Software-Defined Network Optimization Using SOL Mar 2016
Victor Heorhiadi, Michael K. Reiter, Vyas Sekar
(USENIX NSDI 2016 (NSDI '16))

SNIPS: A software-defined approach for scaling intrusion prevention systems via offloading Dec 2014
Victor Heorhiadi, Seyed K. Fayaz, et al.
(ICISS 2014 (ICISS '14))

New opportunities for load balancing in network-wide intrusion detection systems Dec 2012
Victor Heorhiadi, Michael K. Reiter, Vyas Sekar
(ACM CoNeXT 2012 (CoNeXT '12))

Patents

1. Intent-driven network management (US-12155530-B1) – Amazon Technologies, Nov 2023 – Backes, John David, et al.
2. Network change verification based on observed network flows (US-11743122-B1) – Amazon Technologies, Mar 2022 – Bayless, Samuel, et al.
3. Dynamic evaluation and implementation of network mutations (US-11936558-B1) – Amazon Technologies, Dec 2021 – Qian, Baihu, Bashuman Deb, Justin Lin Hsieh, Daniel William Dacosta, Nick Matthews, Anoop Dawani, et al.
4. Network configuration analysis and management (US-12021902-B1) – Amazon Technologies, Dec 2021 – Qian, Baihu, Bashuman Deb, Justin Lin Hsieh, Daniel William Dacosta, Nick Matthews, Viktor Heorhiadi, et al.
5. Failure recovery testing framework for microservice-based applications (US-9842045-B2) – IBM, Feb 2016 – Heorhiadi, Viktor, Hani T. Jamjoom, and Shriram Rajagopalan
6. Determining risk of malware infection in enterprise hosts (US-9674210-B1) – Univ. of North Carolina at Chapel Hill / EMC IP Holding, Nov 2014 – Oprea, Alina M., et al.